

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CLASS

TUTOR'S
NAME

CHEMISTRY

9729/03

Paper 3 Free Response

18 September 2023

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in. Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper. If additional space is required, you should use the pages at the end of this booklet. The question number must be clearly shown.

Section A

Answer all questions.

Section B

Answer **one** question.

A Data Booklet is provided.

The use of an approved scientific calculator is expected, where appropriate.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/20
2	/25
3	/15
4	/20
5	/20
Total	/80

This document consists of **30** printed pages.

[Turn Over

Section A

Answer **all** the questions in this section.

- 1 (a) There is said to be a 'diagonal relationship' between elements of the second and third periods of the Periodic Table, exemplified by lithium and magnesium, and by beryllium and aluminium. This relationship leads to diagonal pairs of elements having similar properties.
- (i) By considering the trends in electronegativity across the period and down the group, suggest why beryllium and aluminium have similar properties.
[1]
- (ii) In terms of structure and bonding, explain why the boiling point of beryllium chloride is much lower than that of beryllium oxide.
[2]
- (iii) Beryllium chloride dissolves in water to give $[\text{Be}(\text{H}_2\text{O})_4]^{2+}$.
Predict and explain, with balanced equations, the effect of adding beryllium chloride to a beaker of litmus solution.
[3]

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- Draw a 'dot-and-cross' diagram of this anion, showing only the valence electrons. In your diagram, use the symbols '•' and '×' to distinguish electrons from different atoms, and the symbol '□' for any additional electrons responsible for the overall negative charge.
- [1]

[illegible]

- (c) Aryl esters can be converted to aryl ketones by a reaction known as the Fries rearrangement.

In the presence of AlCl_3 , the acetyl group, $-\text{COCH}_3$ in phenyl ethanoate, is transferred to the 2-position of the benzene ring to form 2-hydroxyacetophenone. A proposed mechanism for the Fries rearrangement of phenyl ethanoate is shown in Fig. 1.1.

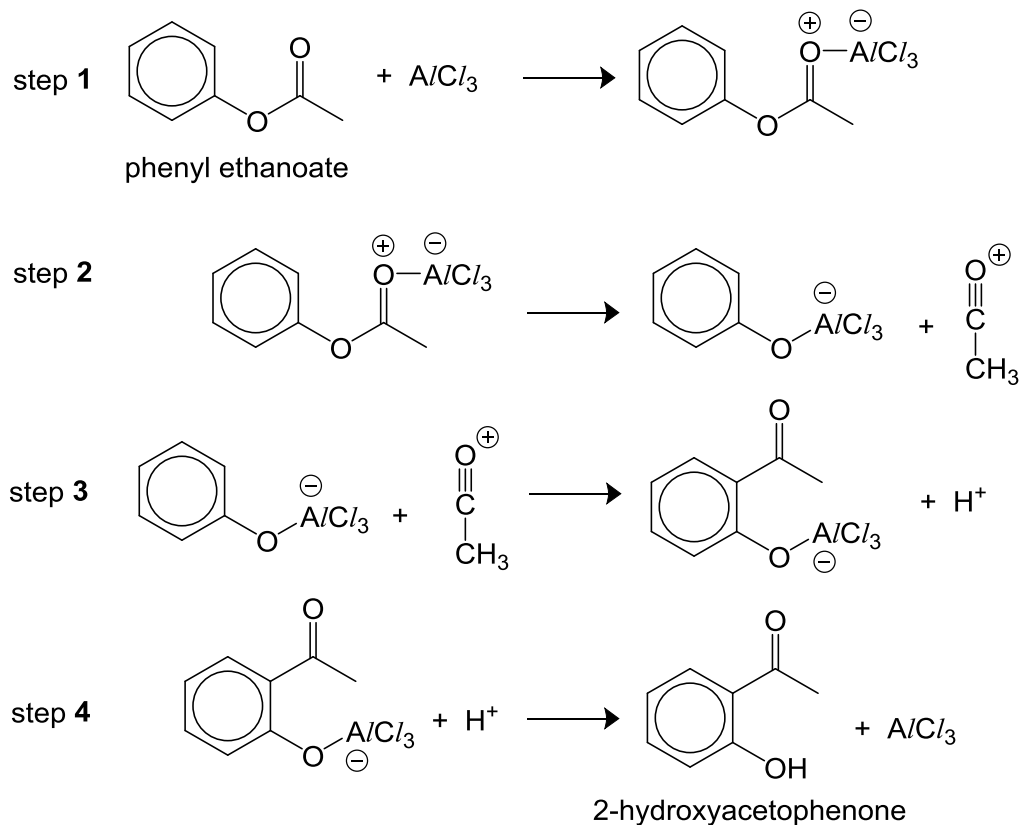


Fig. 1.1

- Write an overall equation for the Fries rearrangement of phenyl ethanoate. [1]
- By considering the mechanism in Fig. 1.1 and your overall equation in (c)(i), state and explain the roles of AlCl_3 in Fries rearrangement. [2]
- Step 3 of the mechanism in Fig. 1.1 occurs in two stages.
 - In stage 1, benzene attacks the carbon atom of CH_3CO^+ to form an intermediate with delocalised positive charge.
 - In stage 2, the loss of a proton restores the aromaticity of the compound.

Draw stages 1 and 2 for this mechanism. Show all relevant charges, dipoles, lone

- CC(=O)Oc1ccccc1C(=O)O

aryl ester **B**

By considering the mechanism in **(c)(iii)**, explain how the rate of Fries rearrangement might differ between phenyl ethanoate and aryl ester **B**.
[2]

[illegible]

2 Mazak is an alloy of magnesium, aluminium, zinc and copper which is widely used to make die-cast toys. Magnesium and aluminium are adjacent elements in the third Period of the Periodic Table; zinc and copper are next to each other in the first set of transition elements.

	metal 1	metal 2	metal 3	metal 4
density / g cm ⁻³	1.74	2.70	7.13	8.94

- Use the *Data Booklet* to identify this residue. Explain your answer. [2]

[illegible]

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- (c) Copper minerals often contain copper(II) sulfide, CuS , mixed with the sulfides of silver and zinc. After roasting in the air to produce the oxides and reduction to the crude metal with carbon, the impure solid is purified by electrolysis.

With reference to the *Data Booklet*, describe the electrode reactions that take place during this electrolysis, and explain in detail how each of the two impurity metals is removed from copper. [3]

[illegible]

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(d) Zinc has some properties that are similar to those of a Group 2 metal and some that are those of a transition element.

(i) Zinc carbonate and magnesium carbonate decompose when heated to give similar products. Write an equation for the thermal decomposition of zinc carbonate. [1]

(ii) Use the *Data Booklet* to deduce whether zinc carbonate will decompose at a higher or lower temperature than magnesium carbonate. Explain your answer. [2]

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(e) The acid-base behaviour of aluminium oxide, Al_2O_3 , shows similarities to that of magnesium oxide, MgO , on the one hand, and phosphorus(V) oxide, P_4O_{10} , on the other.

Describe what these similarities are, and explain why aluminium oxide occupies the in-between position. Write equations for all the reactions you choose to illustrate your answer. [4]

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(f) (i) State how the reactivity of the halogens as oxidising agents varies down the group, and relate this variation to relevant $E^\ominus$ values. [2]

(ii) Describe a reaction that illustrates the relative oxidising abilities of two halogens of your choice. [1]

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[Total: 25]

- 3 (a) Gilman reagents, R_2CuLi where R is an alkyl group, are formed from the reaction between organolithium compound with copper(I) halide.

An example of the synthesis of a Gilman reagent is shown in Fig. 3.1.

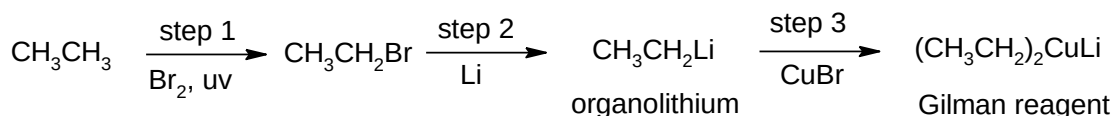


Fig. 3.1

- (i) Describe the mechanism of the reaction in step 1 to form bromoethane. [3]
- (ii) Use data from the *Data Booklet* to calculate the enthalpy change of reaction for step 1. [1]
- (iii) Suggest why the reaction in step 1 continues with an increasing rate after brief exposure to the ultra-violet (UV) light has stopped. [2]
- (iv) Step 2 in Fig. 3.2 is a reduction reaction.

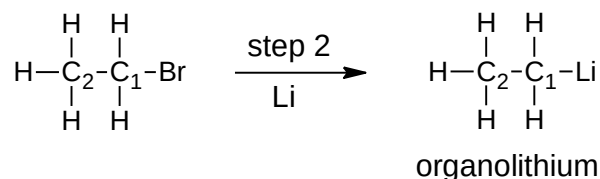


Fig. 3.2

Describe the change in oxidation state of the C_1 atom in step 2. [1]

- (v) Using E° values from the *Data Booklet*, explain why the use of Li to produce CH_3CH_2Li is preferred over Cu in step 2. [1]

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$$\text{CH}_3\text{CH}=\text{CHCOCH}_3 \xrightarrow[\text{H}^+]{(\text{CH}_3\text{CH}_2)_2\text{CuLi}} \text{CH}_3\text{CH}_2\text{CH}_2\text{COCH}_3$$
- 

$$\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 \xrightarrow[\text{H}^+]{(\text{CH}_3\text{CH}_2)_2\text{CuLi}} \text{no reaction}$$

Compound **C**, $C_{10}H_{11}ON$, gives an orange precipitate **D** with 2,4-dinitrophenylhydrazine but does not form a silver mirror with Tollens' reagent. Compound **C** decolourises bromine water to give a white precipitate **E**, $C_{10}H_{11}O_2NBr_2$. On addition of a Gilman reagent, $(CH_3CH_2)_2CuLi$, to **C** followed by careful acidification, compound **F**, $C_{12}H_{17}ON$, is formed. Compound **F** gives an orange precipitate with 2,4-dinitrophenylhydrazine. One mole of compound **F** reacts with one mole of aqueous bromine.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

[Total: 15]

Section B

Answer **one** question from this section.

- 4 (a) Phosphorus pentachloride, PCl_5 , is a solid which melts at 162°C .
- (i) Describe and explain how the entropy of a sample of PCl_5 will change when it is liquefied.
[1]
- (ii) Suggest an observation made when PCl_5 is in contact with water. Include an equation for the reaction that takes place and the approximate pH value of the resulting solution.
[2]

- (b) PCl_5 dissolves in some polar solvents without reaction. Due to the presence of two ions $[\text{PCl}_4]^+$ and $[\text{PCl}_6]^-$, such solutions conduct electricity.

- (i) Draw and name the shape of each of these ions.
[2]

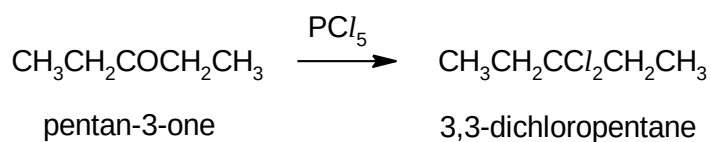
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The presence of $[PCl_4]^+$ and $[PCl_6]^-$ make the reaction between PCl_5 and ketones possible. PCl_5 and ketone reacts in a 1:1 molar ratio to give gem-dichlorides.

A gem-dichloride contains two chlorine atoms bonded to the same carbon atom.



The reaction between pentan-3-one and PCl_5 to form 3,3-dichloropentane involves a series of five steps as shown in Fig. 4.1.

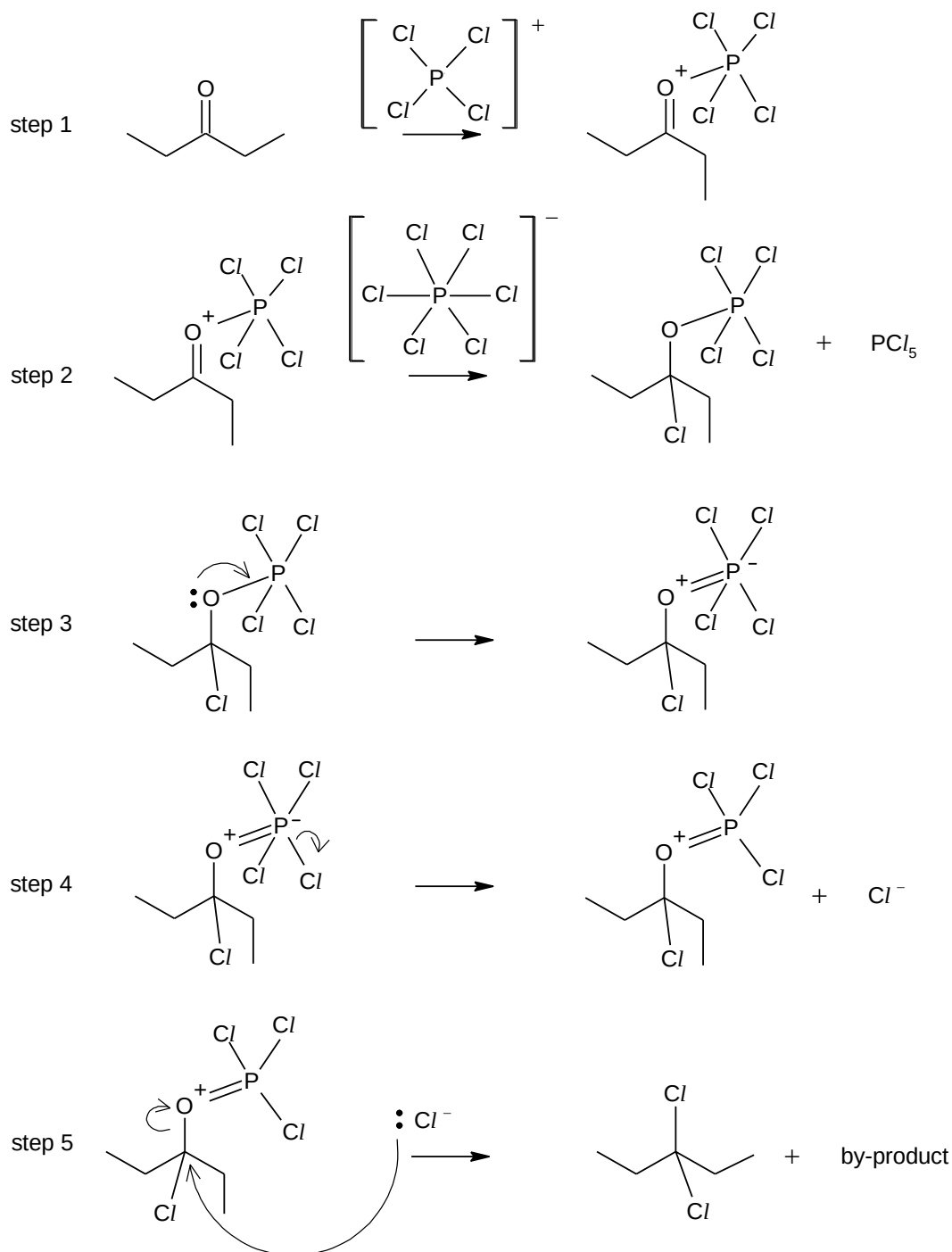


Fig. 4.1

- (ii) Complete Fig. 4.1 to suggest the mechanism for steps 1 and 2. Show all relevant charges, lone pairs and curly arrows. Hence, suggest the phosphorous containing by-product of step 5. [3]
- (iii) Heating 3,3-dichloropentane with ethanolic KOH produces a compound **G**, C_5H_8 , that contains two sp hybridised carbons. Suggest the structure of **G**. [1]
- (iv) Describe, in terms of orbital overlap, the bonding between the two sp hybridised carbon atoms in **G**. Draw a clearly labelled diagram to illustrate your answer. [2]

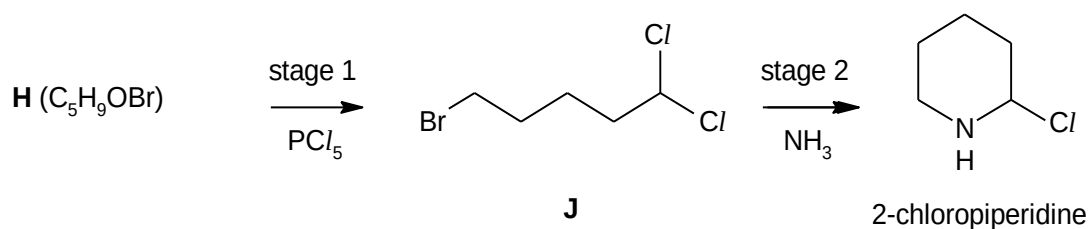
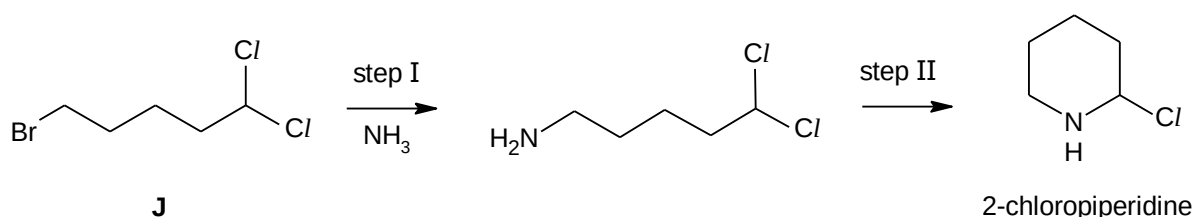


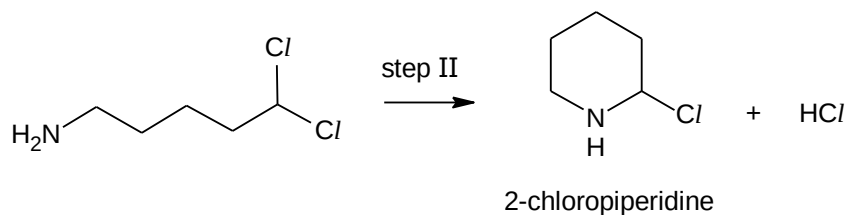
Fig. 4.2

In stage 1, **J** can be synthesised using **H** and PCl_5 via the same five steps described in Fig. 4.1.

In stage 2, limited ammonia is added to **J** to form 2-chloropiperidine. Two substitutions are involved in the formation.



Step II of stage 2 is represented by the following overall equation.



- (i) Suggest the structure of **H**. [1]
- (ii) Use relevant data from the *Data Booklet* to calculate the enthalpy change of reaction for step II of stage 2. [1]
- (iii) 2-chloropiperidine is formed in step II via a pentavalent transition state.

Describe the mechanism for the conversion. Show relevant lone pairs, dipoles, and curly arrows to indicate the movement of electron pairs. [3]

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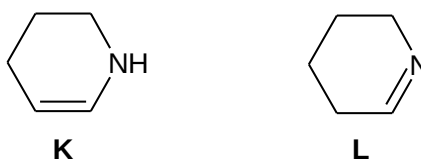
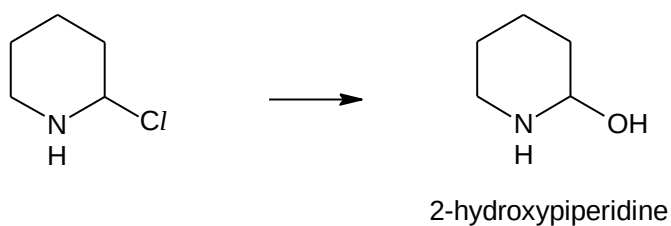
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- Products are formed during the preparation of organic compounds. This usually occurs because the reagents can react in more than one way.

2-hydroxypiperidine can be synthesised using 2-chloropiperidine.



Explain why the two isomers do not show stereoisomerism.
[1]

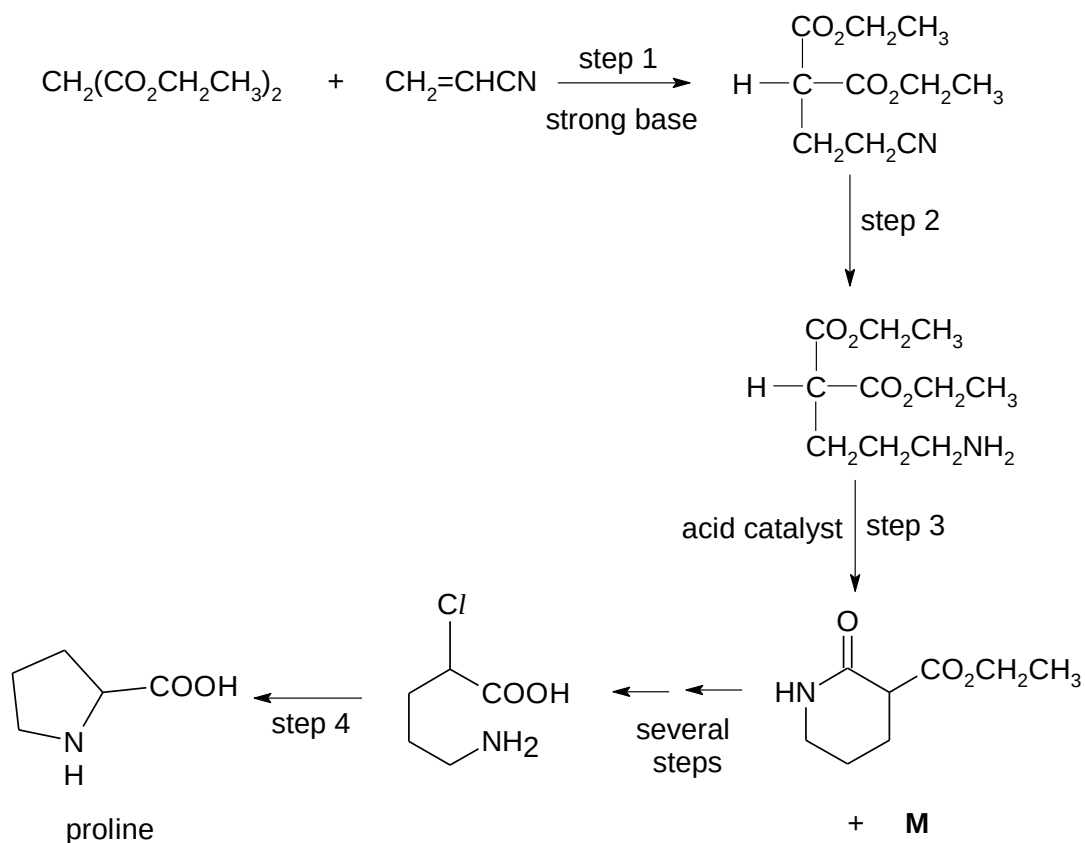
[illegible]

[Total: 20]

- 5 (a)** Describe and explain the trend and variation in first ionisation energy across Period 3. Sketch a diagram to support your answer. [4]

[illegible]

- (b) Proline (Pro) is a naturally occurring amino acid that was first synthesised in the laboratory using a multi-stage synthetic route.



- (i) Proline exhibits stereoisomerism. Draw three-dimensional structures to show the two stereoisomers of proline. [1]
- (ii) Name the type of reaction for steps 1, 3 and 4. [3]
- (iii) State the reagents and conditions required for step 2. [1]
- (iv) Suggest the identity of **M** formed in step 3. [1]

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(c) LiAlH_4 is one of the few reducing agents that can reduce nitrogen containing organic compounds like amides and nitriles. It is highly reactive and has to be handled with care. In the laboratory, excess LiAlH_4 is destroyed as follows.

- Water is added; hydrogen is evolved.
- Dilute acid is added to dissolve the mixture of hydroxides.

(i) Suggest equations for the reactions that occur during the two processes above.
[2]

(ii) Calculate the volume of hydrogen evolved at 25°C and 1 atm if 1.80g of LiAlH_4 were decomposed in this way.
[2]

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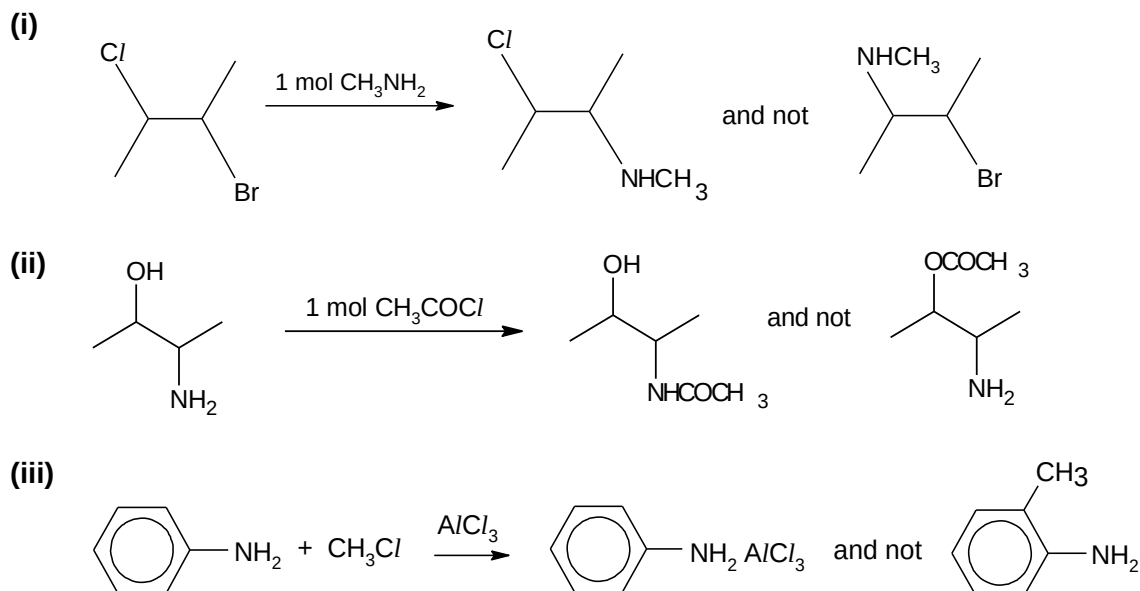
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- (d) Nitrogen compounds react with halogen-containing compounds to give different products, depending on the conditions. For the following reactions, explain the preferential production of the product indicated. [3]



- NC(=O)CNCCS(=O)(=O)O

[3]

[illegible]

Additional answer space

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